

# The Tom Klootwijk Manifold

A formally valid 7-dimensional compact manifold

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) = \prod_{i=0}^6 S^1(r_i)$$

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) = \{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2, i = 0, \dots, 6\}$$

|                        |                                                                  |
|------------------------|------------------------------------------------------------------|
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# 1 Verdict

The source dossier contains a useful creative direction but not a valid theorem as written. Its strongest defensible ingredients are: seven dimensions for seven orthogonal directions; compact procedural notation; rounded rather than jagged visual factors; deterministic personalization; and a GPU-friendly visualization. Its claims that o-indexing changes Hamming distance, that a parity bit can perform the 943-to-937 transition, that  $SDF = 0$  automatically creates a sphere, and that the rendering has zero VRAM or constant frame cost are false.

This specification replaces those claims with a standard, self-contained object: a product of seven circles. The result is a smooth, compact, connected, intrinsically flat 7-manifold with a global orthonormal frame. The chosen personal name is a nomenclature decision for this document; it is not a claim of external peer-review recognition or legal registration.

## One-line definition

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) := \prod_{i=0}^6 S^1(r_i), \quad r_i > 0.$$

## 2 Most compact literal definitions

The following are equivalent descriptions of the same manifold.

### 2.1 Product literal

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) = S^1(r_0) \times \cdots \times S^1(r_6).$$

### 2.2 0-indexed implicit literal

$$F_i(x) = x_{2i}^2 + x_{2i+1}^2 - r_i^2 = 0, \quad i = 0, \dots, 6.$$

Thus  $\mathcal{K}_{\text{TK}}(\mathbf{r}) = F^{-1}(0) \subset \mathbb{R}^{14}$ , with  $F : \mathbb{R}^{14} \rightarrow \mathbb{R}^7$ .

### 2.3 Parametric literal

For  $\theta \in (\mathbb{R}/2\pi\mathbb{Z})^7$  and author phase  $\phi(\alpha)$ ,

$$\Phi_\alpha(\theta)_{2i} = r_i \cos(\theta_i + \phi_i), \quad \Phi_\alpha(\theta)_{2i+1} = r_i \sin(\theta_i + \phi_i).$$

The phase changes only the chosen coordinates/visual instance; it does not alter the manifold's diffeomorphism type.

### 2.4 Metric and orthogonality literal

$$g = \sum_{i=0}^6 r_i^2 d\theta_i^2, \quad E_i = r_i^{-1} \partial_{\theta_i}, \quad g(E_i, E_j) = \delta_{ij}.$$

This is the exact formal statement that there are seven pairwise perpendicular unit tangent directions at every point.

### 2.5 Code literal

```
for (int i = 0; i < 7; ++i) {
  x[2*i]   = r[i] * cos(theta[i] + phase[i]);
  x[2*i+1] = r[i] * sin(theta[i] + phase[i]);
}
```

## 3 Proof of validity

### 3.1 Smoothness and dimension

Define  $F_i(x) = x_{2i}^2 + x_{2i+1}^2 - r_i^2$ . At any  $x \in F^{-1}(0)$ , the  $i$ -th row of  $DF_x$  is zero except in the pair  $(2i, 2i + 1)$ , where it equals  $(2x_{2i}, 2x_{2i+1})$ . Because  $r_i > 0$ , this pair is nonzero. The seven rows have disjoint supports, hence are linearly independent. Therefore  $\text{rank } DF_x = 7$  everywhere on the zero set. By the regular-value theorem,  $F^{-1}(0)$  is a smooth embedded manifold of dimension  $14 - 7 = 7$ .

### 3.2 Global orthonormal frame

The tangent vector  $\partial\Phi/\partial\theta_i$  is nonzero only in coordinate pair  $i$ :

$$\frac{\partial\Phi}{\partial\theta_i} = (0, \dots, -r_i \sin(\theta_i + \phi_i), r_i \cos(\theta_i + \phi_i), \dots, 0).$$

Distinct tangent vectors have disjoint supports and dot product zero; the squared norm of the  $i$ -th vector is  $r_i^2$ . Normalizing gives the seven global orthonormal fields  $E_i$ .

### 3.3 Compactness, topology, and roundness

Each circle is compact and connected, so their finite product is compact and connected. Every factor is round, but the product is a 7-torus, not a sphere. It is intrinsically flat under the product metric even though its embedding in  $\mathbb{R}^{14}$  is extrinsically curved. This distinction supplies a precise version of “round without jaggedness” without pretending that  $\text{SDF} = 0$  creates a hypersphere.

## 4 What was and was not validated from the source

| Claim                               | Verdict              | Reason                                                                     |
|-------------------------------------|----------------------|----------------------------------------------------------------------------|
| Seven perpendicular directions      | Valid in dimension 7 | The frame $E_0, \dots, E_6$ is orthonormal.                                |
| Base-4 values 943 to 937 by one bit | Valid arithmetic     | $973 = 33031_4, 943 = 32233_4, 937 = 32221_4$ .                            |
| o-index/LSB changes that distance   | Invalid              | $943 \oplus 937 = 6 = 110_2$ , so the Hamming distance is 2.               |
| Parity bit executes the mutation    | Invalid              | Relabeling bit positions does not change XOR or Hamming distance.          |
| SDF = 0 makes a sphere              | Invalid              | Parity is a check value, not a general state-transition mechanism.         |
| Procedural storage is compact       | Conditionally valid  | Zero specifies a level set; geometry comes from the field being evaluated. |
| Zero VRAM / $O(1)$ frame cost       | Invalid              | Parameters can be compact, but evaluation still costs work and memory.     |
|                                     |                      | Render targets, code/resources, samples, vertices, and pixels remain.      |

## 5 Personalization without corrupting the mathematics

Let  $\alpha$  be the SHA-256 fingerprint of the private canonical author record. This specification uses

$$\alpha = 7d85cc5019309659b8279866b0bf6dde69ef18373d25dc71051fe8824ee2e3b4.$$

Seven phases and seven 3D projection planes are deterministically derived from  $\alpha$ . The private supplied BSN string is not used as a mathematical premise, and it is not embedded in runtime code. This keeps authorship metadata separate from proof.

| Index | Phase (radians) | Slice frequency ( $m_i, n_i$ ) | Angular speed |
|-------|-----------------|--------------------------------|---------------|
| 0     | 3.080712791     | (1, 0)                         | 0.17          |
| 1     | 5.014583196     | (0, 1)                         | -0.13         |
| 2     | 0.618194258     | (1, 1)                         | 0.11          |
| 3     | 3.690086659     | (1, -1)                        | -0.19         |
| 4     | 4.519778518     | (2, 1)                         | 0.07          |
| 5     | 3.740420404     | (1, 2)                         | 0.23          |
| 6     | 4.338001794     | (2, -1)                        | -0.09         |

## 6 Unity 6.3 implementation

### 6.1 Exact implementation layer

`TKManifoldMath` implements the 14-coordinate embedding, seven implicit residuals, and tangent vectors. Runtime tests verify constraint error and the tangent Gram matrix.

### 6.2 Renderable 3D layer

A screen cannot directly display a 7D manifold. The mesh component evaluates

$$X(u, v, t) = P(\Phi_\alpha(A_t(u, v))),$$

where  $A_t : T^2 \rightarrow T^7$  is the integer-frequency phase slice  $\theta_i = m_i u + n_i v + \omega_i t + \phi_i$ , and  $P : \mathbb{R}^{14} \rightarrow \mathbb{R}^3$  is a fixed linear projection. The render therefore shows a 3D projection of a 2D slice. Self-intersections are allowed, and seven-way orthogonality is not claimed after projection.

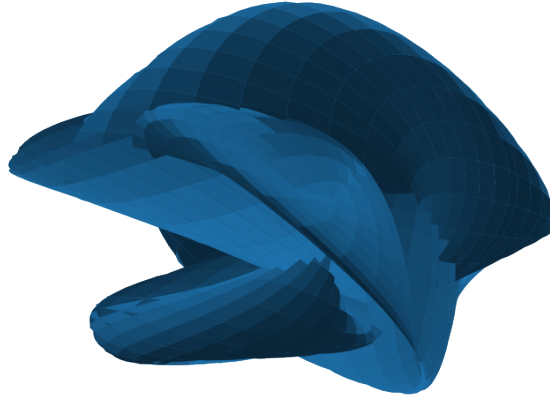


Figure 1: A generated 3D projection of the personalized periodic 2D slice. This is a visualization, not the whole 7D manifold.

### 6.3 Optional SDF surrogate

The package also includes a rounded 3D field made from seven capsule distance functions and a smooth-min union. It is useful as a procedural art object but is explicitly labeled a surrogate. Its per-pixel work is bounded by roughly seven distance evaluations per march step, not constant time.

### 6.4 Default complexity

For  $N_u = 96$  and  $N_v = 48$ , the mesh has  $(N_u + 1)(N_v + 1) = 4,753$  vertices and  $2N_uN_v = 9,216$  triangles. Animated CPU deformation is  $O(7N_uN_v)$  per update and mesh memory is  $O(N_uN_v)$ . The SDF surrogate is  $O(7S)$  field operations per shaded pixel for at most  $S$  march steps, plus normal estimation and framebuffer memory.

### 6.5 Installation

1. Open a Unity 6000.3 URP project; the delivery targets editor 6000.3.22f1 and URP 17.3.0.
2. In Package Manager choose **Add package from disk**. Select **package.json** from the delivered package folder **com.tomklootwijk.manifold**.
3. Use **GameObject > Tom Klootwijk Manifold** and choose **Create Projected 7-Torus Slice**.
4. Optionally create the rounded SDF surrogate; keep its cube scale uniform.
5. Run **Window > General > Test Runner** and execute the package tests.

## 7 Validation record

The independent Python validator reproduced the author fingerprint and checked the base-4 values, XOR result, Hamming distance, 250 random manifold samples, seven-circle residuals, tangent orthogonality and norms, and the personalized projection planes. Numerical errors were at floating-point roundoff scale. Here, formal validation means an explicit definition and complete mathematical proof plus executable checks; the proof was not machine-checked in a theorem prover.

An actual Unity Editor import, C# compilation, shader compilation, and GPU run were **not** executed in the build environment because the Unity Editor was unavailable. The archive includes NUnit runtime tests so the remaining validation occurs immediately

in the target editor rather than being silently asserted here.

## 8 Nomenclature and privacy status

“The Tom Kloonwijk Manifold” is the author’s selected label for the object defined in this document. The full supplied BSN string appears only in the private copy and private author record. The public runtime package uses only the fingerprint. Keep this private copy out of public repositories, issue trackers, screenshots, and asset-store submissions.

## 9 References and implementation basis

- Supplied source dossier, especially its proposed name/notation, its 7D discussion, and its internal critique of geometry, topology, and information theory.
- Unity 6.3 User Manual and Unity 6000.3.22fi release documentation.
- Universal Render Pipeline 17.3.0 package documentation; the package manifest pins this version for the Unity 6000.3 target.
- Standard regular-value theorem and product-manifold construction; the proof needed here is given in full above.

## A Literal definition card

$$\begin{aligned}
 \mathcal{K}_{\text{TK}}(\mathbf{r}) &= \prod_{i=0}^6 S^1(r_i) \\
 &= \{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2, i = 0, \dots, 6\}, \\
 \Phi_\alpha(\theta)_{2i:2i+1} &= r_i(\cos(\theta_i + \phi_i), \sin(\theta_i + \phi_i)), \\
 g &= \sum_{i=0}^6 r_i^2 d\theta_i^2, \quad g(E_i, E_j) = \delta_{ij}, \\
 X(u, v, t) &= P\Phi_\alpha((m_i u + n_i v + \omega_i t)_{i=0}^6).
 \end{aligned}$$

## B Private/public author-record note

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# The Tom Klootwijk Manifold

## Spatiotemporal SDF Addendum

A rigorous compact definition of the “spatiotemporal SDF  $o(1)$  geometrical topological substrate”

Let  $y = Q(t)^T(x - c(t))$  and, for  $i = 0, \dots, 6$ ,

$$\Delta_i(x, t) = \|y_{2i:2i+1}\|_2 - r_i(t), \quad \mathcal{D}_\tau(x, t) = \|\Delta(x, t)\|_2 - \tau(t).$$

The compact substrate is

$$\mathfrak{S}_{\text{TK}}^{\text{ST}} := (\mathcal{W}_{\text{TK}} = \Delta^{-1}(0), \Sigma_{\text{TK}, \tau} = \mathcal{D}_\tau^{-1}(0), \{\widehat{\mathcal{D}}_h\}_{h>0}), \quad \|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1).$$

|                          |                                                                  |
|--------------------------|------------------------------------------------------------------|
| Author                   | Tom Klootwijk                                                    |
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# 1 Status and scope

This addendum does not replace the original definition

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) = \prod_{i=0}^6 S^1(r_i) \subset \mathbb{R}^{14}.$$

It adds a mathematically controlled time parameter, an exact distance to that codimension-seven core, and a scalar signed-distance shell that has a genuine inside, boundary, and outside.

The supplied addendum dossier correctly centers computer-graphics SDF notation on sign and the zero-level boundary. It then proposes an Eikonal/topological “hinge” and a chronotemporal use of little- $o$  notation, but it also mixes those ideas with unsupported quantum-phase, parity-bit, homogeneous-time, and zero-cost-rendering claims. This addendum retains the geometric intent and removes those unsupported identifications.

## Formal verdict

The phrase “spatiotemporal SDF  $o(1)$  geometrical topological substrate” becomes rigorous only after four distinctions:

1. the codimension-seven core uses a *vector* normal-distance field;
2. a scalar signed distance is assigned to the boundary of a regular tube around that core;
3. time is an independent parameter and the moving zero set is a worldvolume;
4. lowercase  $o(1)$  describes vanishing approximation error, not execution time.

# 2 Canonical compact definition

Let  $I \subset \mathbb{R}$  be a time interval. Choose  $C^2$  data

$$Q : I \rightarrow SO(14), \quad c : I \rightarrow \mathbb{R}^{14}, \quad r_i : I \rightarrow (0, \infty), \quad \tau : I \rightarrow (0, \infty)$$

with the uniform regularity guard

$$r_* := \inf_{t \in I, 0 \leq i \leq 6} r_i(t) > 0, \quad 0 < \tau(t) < r_* \quad \text{for all } t \in I.$$

For  $(x, t) \in \mathbb{R}^{14} \times I$ , write  $y = Q(t)^T(x - c(t))$  and define

$$\begin{aligned} \Delta_i(x, t) &:= \sqrt{y_{2i}^2 + y_{2i+1}^2} - r_i(t), \quad i = 0, \dots, 6, \\ \delta(x, t) &:= \|\Delta(x, t)\|_2, \\ \mathcal{D}_\tau(x, t) &:= \delta(x, t) - \tau(t). \end{aligned}$$

Then

$$\mathcal{W}_{\text{TK}} := \{(x, t) : \Delta(x, t) = 0\}, \quad \Sigma_{\text{TK}, \tau} := \{(x, t) : \mathcal{D}_\tau(x, t) = 0\}.$$

The numerical consistency family  $\{\widehat{\mathcal{D}}_h\}_{h>0}$  must satisfy, for every compact regular band  $B$  around the shell,

$$\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1) \quad \text{as } h \downarrow 0.$$

This triple is the addendum’s formal substrate  $\mathfrak{S}_{\text{TK}}^{\text{ST}}$ .

## 2.1 Shortest literal forms

|                          |                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| <b>Vector field:</b>     | $\Delta_i = \ [Q^T(x - c)]_{2i:2i+1}\ _2 - r_i(t).$                                   |
| <b>Scalar shell SDF:</b> | $\mathcal{D}_\tau = \ \Delta\ _2 - \tau(t).$                                          |
| <b>Core topology:</b>    | $\mathcal{W}_{\text{TK}} \cong \mathbb{T}^7 \times I.$                                |
| <b>Shell topology:</b>   | $\Sigma_{\text{TK}, \tau} \cong \mathbb{T}^7 \times S^6 \times I.$                    |
| <b>Geometry:</b>         | $\ \nabla_x \mathcal{D}_\tau\ _2 = 1$ on the regular shell.                           |
| <b>Motion:</b>           | $V_n = -\partial_t \mathcal{D}_\tau$ on $\mathcal{D}_\tau = 0.$                       |
| <b>Consistency:</b>      | $\ \widehat{\mathcal{D}}_h - \mathcal{D}_\tau\ _\infty = o(1), \quad h \downarrow 0.$ |

### 3 Why this is an exact distance construction

#### 3.1 Exact distance to the core

At fixed time, rigid motion by  $Q(t)$  and  $c(t)$  preserves Euclidean distance. In each orthogonal coordinate pair, the distance from  $y_{2i:2i+1} \in \mathbb{R}^2$  to the circle of radius  $r_i(t)$  is

$$\left| \sqrt{y_{2i}^2 + y_{2i+1}^2} - r_i(t) \right| = |\Delta_i(x, t)|.$$

The product minimization separates across the seven orthogonal planes, so

$$\boxed{\text{dist}(x, \mathcal{K}_{\text{TK},t})^2 = \sum_{i=0}^6 \Delta_i(x, t)^2}, \quad \boxed{\delta(x, t) = \text{dist}(x, \mathcal{K}_{\text{TK},t})}.$$

Thus  $\delta$  is not merely a heuristic distance estimator: it is the exact Euclidean distance to the time slice  $\mathcal{K}_{\text{TK},t}$ .

#### 3.2 Why the core itself is not a scalar signed surface

The embedded  $\mathcal{K}_{\text{TK},t} \subset \mathbb{R}^{14}$  has codimension seven. It does not divide the ambient space into a canonical inside and outside, so a single scalar sign on  $\delta$  would be artificial. The sign-bearing object is instead the codimension-one tube boundary

$$\Sigma_t = \{x : \delta(x, t) = \tau(t)\} = \partial\{x : \delta(x, t) \leq \tau(t)\}.$$

On the regular tube,  $\mathcal{D}_\tau < 0$  inside,  $\mathcal{D}_\tau = 0$  on the boundary, and  $\mathcal{D}_\tau > 0$  outside.

#### 3.3 Eikonal identity

The seven gradients  $\nabla_x \Delta_i$  are pairwise orthonormal wherever the coordinate-pair radii are nonzero. On  $\mathcal{D}_\tau = 0$ , the guard  $\tau < r_*$  implies every pair radius is positive and  $\Delta \neq 0$ . Therefore

$$\nabla_x \mathcal{D}_\tau = \sum_{i=0}^6 \frac{\Delta_i}{\|\Delta\|_2} \nabla_x \Delta_i, \quad \boxed{\|\nabla_x \mathcal{D}_\tau\|_2^2 = \sum_{i=0}^6 \frac{\Delta_i^2}{\|\Delta\|_2^2} = 1}.$$

This is the precise Eikonal statement. It holds in the regular tubular band; it is not a claim that every arbitrary implicit field is an exact SDF everywhere.

## 4 Topology and the spatiotemporal worldvolume

#### 4.1 Core topology

A direct parameterization is

$$\Phi(\theta, t) = c(t) + Q(t)(r_i(t)(\cos \theta_i, \sin \theta_i))_{i=0}^6.$$

Because every  $r_i(t) > 0$  and  $Q(t)$  is orthogonal,  $\Phi$  is a smooth embedding for each time. Hence

$$\mathcal{K}_{\text{TK},t} \cong \mathbb{T}^7, \quad \boxed{\mathcal{W}_{\text{TK}} \cong \mathbb{T}^7 \times I},$$

with spatial dimension 7 and spacetime dimension 8.

#### 4.2 Shell topology

For  $u = (u_0, \dots, u_6) \in S^6$ , define

$$\Psi(\theta, u, t) = c(t) + Q(t)((r_i(t) + \tau(t)u_i)(\cos \theta_i, \sin \theta_i))_{i=0}^6.$$

The guard  $\tau < r_*$  makes every factor radius positive. Conversely, each shell point uniquely determines its seven angles and  $u = \Delta/\tau$ . Therefore  $\Psi$  is a diffeomorphism and

$$\Sigma_t \cong \mathbb{T}^7 \times S^6, \quad \boxed{\Sigma_{\text{TK},\tau} \cong \mathbb{T}^7 \times S^6 \times I}.$$

The shell has spatial dimension  $7 + 6 = 13$  and spacetime dimension 14.

### 4.3 What “chronotemporal latch” can rigorously mean

The moving shell is the zero set of one  $C^1$  function on  $\mathbb{R}^{14} \times I$ . Since  $\nabla_x \mathcal{D}_\tau \neq 0$  on that zero set, its outward normal speed is

$$V_n = -\frac{\partial_t \mathcal{D}_\tau}{\|\nabla_x \mathcal{D}_\tau\|_2} = -\partial_t \mathcal{D}_\tau.$$

This equation links consecutive time slices without treating time as a homogeneous graphics coordinate and without any parity-bit mechanism. Topology remains fixed because the explicit diffeomorphism above persists. A topology change can occur only after a guard fails, for example  $r_i \rightarrow 0$ ,  $\tau \rightarrow 0$ ,  $\tau \geq \min_i r_i$ , discontinuous data, or a singular level set.

## 5 The exact meaning of lowercase $o(1)$

Lowercase  $o(1)$  is asymptotic notation. It means a quantity tends to zero. In this addendum,  $h$  is a numerical scale such as a time step, voxel spacing, mesh spacing, or finite-difference step, and

$$\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1)$$

means the implemented field converges uniformly to the exact field on each compact regular band.

For temporal stepping, smoothness gives

$$\mathcal{D}_\tau(x, t+h) = \mathcal{D}_\tau(x, t) + h \partial_t \mathcal{D}_\tau(x, t) + R(x, t, h), \quad \frac{R(x, t, h)}{h} = o(1).$$

The validator uses the deterministic profile included in the Unity package and observes successive values of  $|R(h)|/h$  decreasing by approximately one half when  $h$  is halved, consistent with a second-order Taylor remainder.

If  $\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_\infty \leq \varepsilon_h$  and  $\varepsilon_h \rightarrow 0$ , the exact signed-distance property implies that approximate zeros lie within  $\varepsilon_h$  of the true shell. With continuity and a normal-band sign test, the reverse inclusion is also controlled, so the zero sets converge geometrically.

#### Complexity distinction

The exact field evaluates seven pair lengths and one seven-vector length. Because seven is fixed, a single query is  $O(7) = O(1)$  with respect to variable scene size or mesh resolution. It is never  $o(1)$  runtime. Rendering  $P$  pixels with at most  $S$  sphere-tracing steps is  $O(PS)$  field queries and uses nonzero render-target, code, and resource memory.

## 6 Unity 6.3 implementation

The delivered UPM package is `com.tomklootwijk.manifold` version `1.1.0`, targeting the Unity `6000.3` LTS API line and URP `17.3.0`.

### 6.1 Exact high-dimensional API

`TKSpacetimeSubstrateMath` implements:

- core embedding and shell embedding in  $\mathbb{R}^{14}$ ;
- vector normal coordinates  $\Delta \mapsto \mathbb{R}^7$ ;
- exact core distance  $\delta = \|\Delta\|_2$ ;
- regular tubular SDF  $\mathcal{D}_\tau = \delta - \tau$ ;
- analytic spatial gradient and Eikonal verification;
- identity-frame temporal derivative and  $V_n = -\partial_t \mathcal{D}_\tau$ ;
- regularity guards and optional row-major  $14 \times 14$  orthogonal transforms.

### 6.2 Smooth deterministic motion profile

`TKSpacetimeProfile` supplies smooth center, radius, and tube-radius functions together with their derivatives. Its conservative minimum factor radius is 0.84832, its maximum tube radius is 0.208, and its regularity margin is therefore 0.64032.

### 6.3 Exact 3D SDF witness

The full shell lives in fourteen-dimensional spacetime and cannot be displayed directly. The package therefore includes a dimension-reduced exact ring-torus witness:

$$d_3(p, t) = \left\| \left( \sqrt{p_x^2 + p_z^2} - R(t), p_y \right) \right\|_2 - a(t), \quad 0 < a(t) < R(t).$$

`TKSpacetimeTorusSdfWitness` drives the URP shader and enforces the guard for all animation phases. The witness has a genuine 3D SDF at each time, but it is not mislabeled as the complete  $\mathbb{T}^7 \times S^6$  shell.

### 6.4 Installation and use

1. Open a Unity 6000.3 URP project.
2. In Package Manager choose **Add package from disk** and select the delivered `package.json`, or install the delivered `.tgz`.
3. Use **GameObject > Tom Klootwijk Manifold > Create Spatiotemporal Torus SDF Witness**.
4. Keep the generated cube uniformly scaled.
5. Run the package Runtime tests in the Unity Test Runner.

## 7 Validation record

The standalone mathematical validator passed 500 randomized samples. Its maximum measured errors were:

| Check                                            | Maximum absolute error |
|--------------------------------------------------|------------------------|
| Core normal-coordinate norm                      | $3.69 \times 10^{-16}$ |
| Exact distance versus nearest-point construction | $6.67 \times 10^{-16}$ |
| Shell SDF residual                               | $4.72 \times 10^{-16}$ |
| Eikonal norm error                               | $2.23 \times 10^{-16}$ |

The little- $o$  temporal test used  $h = 10^{-2}, 5 \cdot 10^{-3}, 2.5 \cdot 10^{-3}, 1.25 \cdot 10^{-3}$ . Successive ratios of  $|R(h)|/h$  were approximately 0.50018, 0.50009, and 0.50005.

Static package validation passed JSON/assembly parsing, required-file checks, delimiter balance for all C# and shader files, formula-presence tests, and confirmation that the full supplied BSN and birth date do not appear anywhere in the runtime package.

An actual Unity Editor import, C# compilation by Unity, URP shader compilation, and GPU execution were **not executed** because the Unity Editor is unavailable in the build environment. The delivery includes NUnit tests for that final target-environment validation.

## 8 Source claims retained, narrowed, or rejected

| Source idea                             | Addendum verdict           | Serious replacement                                                                                             |
|-----------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| SDF sign separates inside/outside       | Retained for a shell       | The codimension-one tube boundary has $\mathcal{D}_\tau < 0$ , $= 0$ , and $> 0$ .                              |
| Zero set defines the surface            | Retained with conditions   | Geometry comes from the explicit field; zero alone does not determine a sphere.                                 |
| Eikonal equation                        | Proved in the regular band | $\ \nabla_x \mathcal{D}_\tau\  = 1$ follows from seven orthonormal normal coordinates.                          |
| Zero set permanently fixes topology     | Narrowed                   | Topology is fixed only while the regularity guard and smooth data persist.                                      |
| Hadamard sign basis gives quantum phase | Rejected                   | No quantum state or phase follows from the sign of a classical SDF.                                             |
| Time is homogeneous coordinate $w$      | Rejected                   | Time is the independent parameter $t \in I$ .                                                                   |
| Parity/LSB flip links past and future   | Rejected                   | Temporal evolution is governed by $\partial_t \mathcal{D}_\tau$ ; Hamming distance remains unchanged.           |
| $o(1)$ means constant-time rendering    | Rejected                   | $o(1)$ is vanishing error; one field query is fixed-size $O(1)$ , while rendering scales with pixels and steps. |

## 9 Nomenclature, attribution, and privacy

“The Tom Klootwijk Manifold” and “Tom Klootwijk Spatiotemporal SDF Substrate” are the author’s chosen names in this delivery. The names identify the specification edition and do not by themselves establish external academic recognition, priority, trademark, patent status, or legal ownership.

The private author record is used only to create the SHA-256 fingerprint shown on the title page. Runtime source contains the fingerprint, author name, public document identifiers, and occasion times, but not the supplied BSN or birth date.

## 10 References and implementation basis

- Supplied original manifold dossier and supplied SDF/chronotemporal addendum dossier.
- Unity 6.3 LTS Manual: <https://docs.unity3d.com/6000.3/Documentation/Manual/UnityManual.html>
- URP 17.3.0 documentation: <https://docs.unity3d.com/Packages/com.unity.render-pipelines.universal@17.3/manual/>
- Unity Package Manager package layout: <https://docs.unity3d.com/Manual/cus-layout.html>
- Standard product-manifold, tubular-neighborhood, level-set, and signed-distance constructions; all claims needed for this specific object are proved above.

## A Implementation literal

```
// Fixed 0-indexed factor count: 7.
for (int i = 0; i < 7; ++i) {
    int j = 2 * i;
    delta[i] = hypot(local[j], local[j + 1]) - radius[i];
}
double coreDistance = norm(delta);
double tubularSdf = coreDistance - tubeRadius;
// Consistency requirement: sup |approx_h - tubularSdf| = o(1), h -> 0.
```

## B Compact theorem statement

Assume  $Q, c, r_i, \tau$  are  $C^2$ ,  $Q(t) \in SO(14)$ , and  $0 < \tau(t) < \inf_i r_i(t)$ . Then  $\mathcal{W}_{\text{TK}}$  is a smooth 8-manifold diffeomorphic to  $\mathbb{T}^7 \times I$ ;  $\Sigma_{\text{TK}, \tau}$  is a smooth 14-manifold diffeomorphic to  $\mathbb{T}^7 \times S^6 \times I$ ;  $\mathcal{D}_\tau$  is the exact signed distance to the shell in its regular normal band;  $\|\nabla_x \mathcal{D}_\tau\| = 1$  there; and its normal velocity is  $V_n = -\partial_t \mathcal{D}_\tau$ . Any numerical family satisfying  $\|\hat{\mathcal{D}}_h - \mathcal{D}_\tau\|_\infty = o(1)$  is a consistent approximation of this substrate.